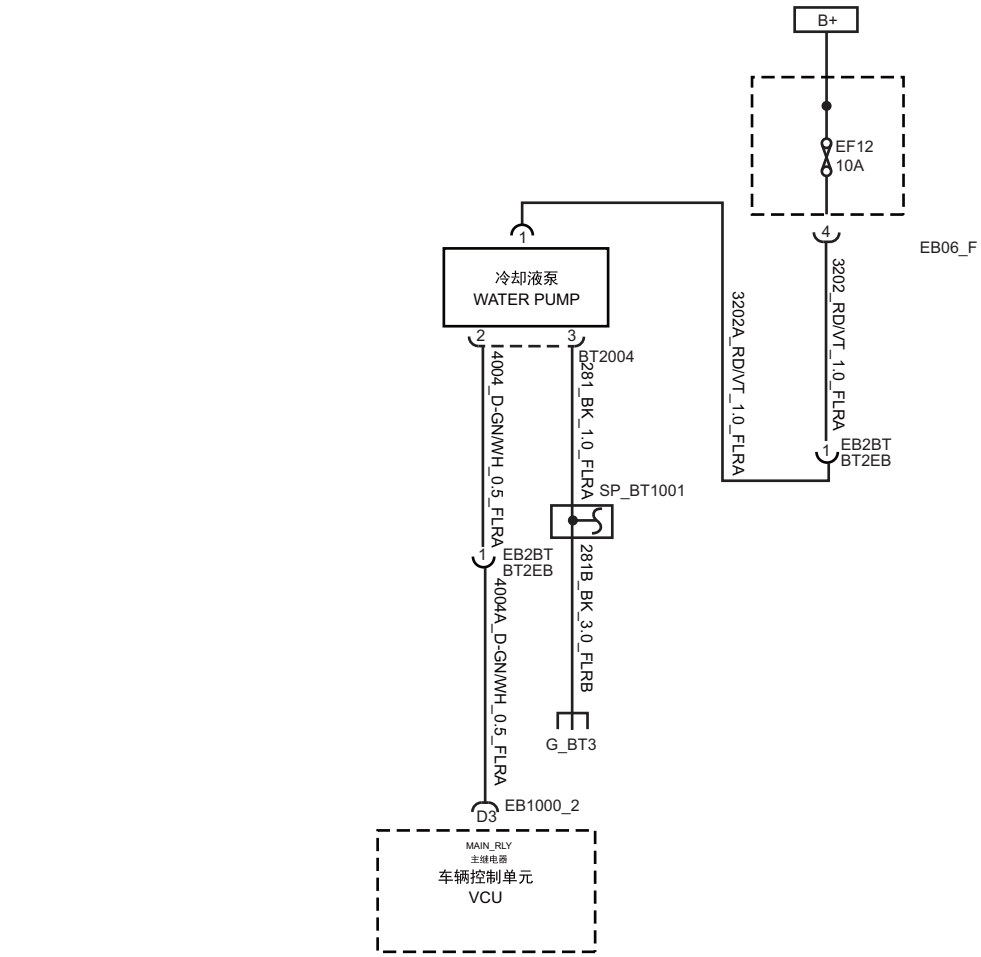
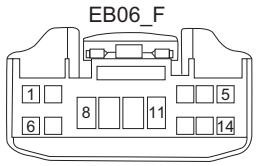
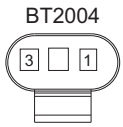
P0A07、 P0A06、 P0A05

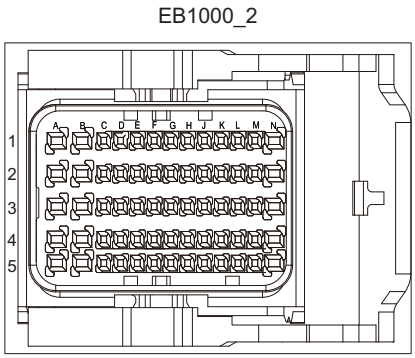
Fault diagnosis code
P0A07：DEM_stWtrPmpSBCfmd
P0A06：DEM_stWtrPmpSGCfmd
P0A05：DEM_stWtrPmpOCCfmd
Detection tools
1 Multimeter and diagnostic apparatus
Possible causes
• Circuit,terminal and adapting piece fault • VCU module fault • The motor controller cooling water pump is faulty.
Conditions for fault code operation
Turn the ignition switch to ON position.
Conditions for fault code setting
P0A07
The underlying diagnosis
P0A06
The underlying diagnosis
P0A05
The underlying diagnosis
To set the effect of a fault code condition
Description of circuit diagram
The VCU module monitors motor controller cooling water pump via low level reference circuit and signal circuit.

Circuit diagram



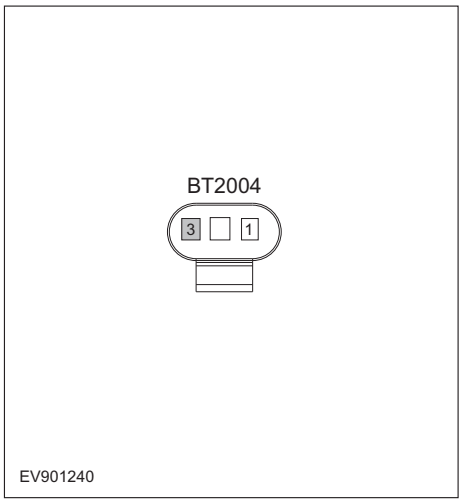


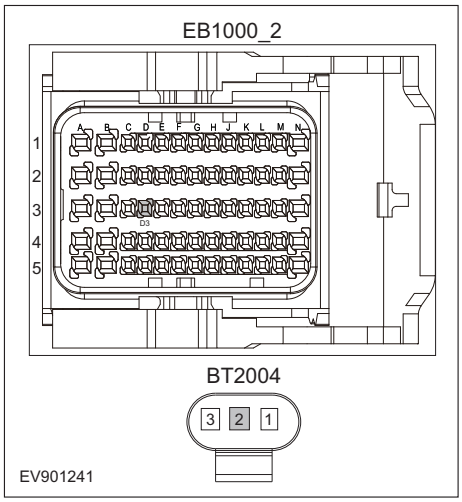




EV901208

<table>
<tr><td colspan="3">Troubleshooting steps</td></tr>
<tr><td>Step 1</td><td colspan="2">General inspection.</td></tr>
<tr><td colspan="2"></td><td>1 Check the fuse EF13 for damage.
2 Check battery capacity.
3 Check the harness plug of motor controller cooling water pump, VCU for damage, poor contact, aging and looseness.
4 Check whether the result is normal or not?</td></tr>
<tr><td colspan="2"></td><td>NO → Repair or replace the faulted parts.</td></tr>
<tr><td colspan="3">YES ↓</td></tr>
<tr><td>Step 2</td><td colspan="2">Check fault codes.</td></tr>
<tr><td colspan="2"></td><td>1 Connect the fault diagnosis instrument to the diagnosis interface.
2 Turn the ignition switch to ON position.
3 Diagnose the VCU with diagnostic apparatus.
4 Is there any other fault code except for P0A07/P0A06/P0A05?</td></tr>
<tr><td colspan="2"></td><td>YES → Refer to:DTC Summary list(VCU).</td></tr>
<tr><td colspan="3">NO ↓</td></tr>
<tr><td>Step 3</td><td colspan="2">Check the supply power of motor controller cooling water pump when it is started or operating.</td></tr>
<tr><td colspan="2"> 
</td><td>1 Turn the ignition switch to OFF.
2 Disconnect the harness plug BT2004 of motor controller cooling water pump.
3 Turn the ignition switch to ON.
4 Measure the voltage between the plug terminal of motor controller cooling water pump and ground.
Measuring circuit: voltage between terminal 1 on plug BT2004 and ground.
Standard value:10~14V
5 Check whether the result is normal or not?</td></tr>
<tr><td colspan="2"></td><td>NO → Repair the circuit.</td></tr>
<tr><td colspan="3">YES ↓</td></tr>
</table>

Step 4	Check the circuit of temperature sensor at the water inlet of the motor controller(open circuit).
	1 Turn the ignition switch to OFF. 2 Measure the voltage between the plug terminal of motor controller cooling water pump and ground. Measuring circuit: resistance between terminal 3 on plug BT2004 and ground. **Standard value: ≈ 0Ω** 3 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 5	Check the circuit of temperature sensor at the water inlet of the motor controller (open circuit).
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug BT2004 of motor controller cooling water pump. 3 Disconnect the harness plug EB1000_2 of GAW module. 4 Measure the resistance between the harness terminal of motor controller cooling water pump and the VCU. Measuring circuit: resistance between terminal 2 on plug BT2004 and terminal D3 on plug EB1000_2. **Standard value: ≈ 0Ω** 5 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 6	Check the circuit of temperature sensor at the water inlet of the motor controller(short circuit).
	1 Turn the ignition switch to OFF. 2 Measure the resistance between the plug terminal of motor controller cooling water pump and ground. Measuring circuit: resistance between terminal 1 on plug BT2004 and ground. Measuring circuit: resistance between terminal 2 on plug BT2004 and ground. **Standard value: ∞** 3 Measure the voltage between the plug terminal of motor controller cooling water pump and ground. Measuring circuit: voltage between terminal 2 on plug BT2004 and ground. Measuring circuit: voltage between terminal 3 on plug BT2004 and ground. **Standard value: ≈ 0V** 4 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	
Step 7	Check the circuit of motor controller cooling water pump(short circuit).
	1 Measure the voltage between the plug terminal of motor controller cooling water pump and ground. Measuring circuit: resistance between terminal 2 and terminal 3 on plug BT2004. **Standard value:** 2 Measure the voltage between the plug terminal of motor controller cooling water pump and the housing. Measuring circuit: resistance between terminal 1 on plug BT2004 and the housing. Measuring circuit: resistance between terminal 2 on plug BT2004 and the housing. Measuring circuit: resistance between terminal 3 on plug BT2004 and the housing. **Standard value: ∞** 3 Check whether the result is normal or not?
	NO → Replace the motor controller cooling water pump.
YES ↓	

Step 8	Test whether VCU is norm.
	1 Turn the ignition switch to OFF. 2 Disconnect and check the harness plug EB1000_1, IP55_A of VCU. 3 Check and repair. • Corrosion(install new adapting piece or terminal-cleaning module and pin). • Pin damaged or bent-install new terminal/pin. • Fly out of pins-install new pins as needed. 4 Reconnect the VCU plug and all other previously disconnected plug. Make sure the pin is properly positioned and fixed. 5 Run the system and determine if the problem persists?
	YES → Change the VCU Module
NO ↓	
At this point the system is functioning normally. The problem may be caused by module connections. Address the root cause of any connector or pin problems.	
Next Step ↓	
Diagnosis end.	